Taming Multiparticle Entanglement

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Paper Information

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Key Contributions:

- Efficient criterion for **genuine multipartite entanglement (GME)** via **PPT mixtures**
- **Semidefinite programming (SDP)** formulation for practical detection
- **Fully decomposable entanglement witnesses**
- **Entanglement Quantification**

Outline

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Motivation

The Challenge: How do we efficiently verify that a state has *genuine* multipartite entanglement that cannot be reduced to bipartite states?

Separability and Entanglement

A bipartite state ρ is **separable** if:

$$|\psi\rangle = \sum_i \lambda_i |u_i\rangle_A \otimes |v_i\rangle_B$$

$$\rho = \sum_i p_i \rho_i^A \otimes \rho_i^B$$

where $\sum_i p_i = 1$ and $p_i > 0$.

Problem: Directly determining separability is **NP-hard**!

- How should we split the system into subsystems?
- It is a costly operation!

[Motivation] Examples of Multiparticle Entangled States

GHZ State (Greenberger-Horne-Zeilinger)

$$|GHZ_n\rangle = \frac{1}{\sqrt{2}}(|00\cdots0\rangle + |11\cdots1\rangle)$$

W State

$$|W_3\rangle = \frac{1}{\sqrt{3}}(|001\rangle + |010\rangle + |100\rangle)$$

Cluster State (4-qubit linear)

$$|Cl_4\rangle = \frac{1}{2}(|0000\rangle + |0011\rangle + |1100\rangle - |1111\rangle)$$

[Motivation] Verification of Entangled States

For pure states, SVD is the most effective way to quantify entanglement. For example, consider the two-qubit Bell state (though this is a trivial example).

Step 1. Write the state. $|\Phi^+\rangle = \frac{1}{\sqrt{2}}|00\rangle + 0|01\rangle + 0|10\rangle + \frac{1}{\sqrt{2}}|11\rangle = \sum_{i,j} c_{ij}|i\rangle_A|j\rangle_B$

Step 2. Form Coefficient Matrix.

$$C = \begin{pmatrix} c_{00} & c_{01} \\ c_{10} & c_{11} \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Step 3. Find SVD of C, that is, $C = U\Sigma V^\dagger$.

$$C = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{1}{\sqrt{2}} \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Step 4. Two Schmidt coefficients $\Rightarrow$ entangled state. Try the other Bell states!

[Motivation] Challenges for Entanglement Detection: Pure States

Exponential cost is inevitable. For an $m \times n$ matrix:

Resource	**Complexity**
Time	$\mathcal{O}(\min(mn^2, m^2n)) \simeq \mathcal{O}(mn \cdot \min(m, n))$
Memory	$\mathcal{O}(mn)$ for matrix + $\mathcal{O}(m^2 + n^2)$ for U, V

For N qubits with bipartition $k|(N-k)$, the matrix size becomes $2^k \times 2^{N-k}$. Things get worse if we must try every possible bipartition, of which there are 2^{N-1}.

[Motivation] Challenges for Entanglement Detection: Mixed States

Even worse, for mixed states there is no "canonical form" like the Schmidt decomposition. The same ρ can be written as:

$$\rho = \sum_k p_k |\psi_k\rangle\langle\psi_k| = \sum_j q_j |\phi_j\rangle\langle\phi_j|$$

with completely different ensembles! One might use only product states (separable), another might not (entangled representation), while the **physical state is the same**.

[Definition] Biseparable States, Genuine Multipartite Entanglement

A state is separable with respect to the bipartition $M|\bar{M}$ if:

$$\rho^{\text{sep}}_{M|\bar{M}} = \sum_i p_i \, \rho^{(i)}_M \otimes \rho^{(i)}_{\bar{M}}$$

A state is **biseparable** if it can be written as a mixture:

$$\rho^{\text{bs}} = \sum_i p_i \rho^{\text{sep}}_{M_i|\bar{M}_i}$$

where $\sum_i p_i = 1$. Note that a non-trivial form (with at least two $p_i \neq 0$) must be a mixed state.

Genuine Multipartite Entanglement (GME)

A state is **genuinely multipartite entangled** if and only if it is **NOT biseparable**.

$$\text{GME} = \text{NOT biseparable}$$

[Example] Biseparable States

Consider a three-qubit system (labeled A, B, C). There are three possible bipartitions: $A|BC$, $B|CA$, and $C|AB$, i.e., $M|\bar{M} \in \{A|BC, B|CA, C|AB\}$. A biseparable state can be written as:

$$\rho^{\text{bs}} = p_{A|BC}\rho^{\text{sep}}_{A|BC} + p_{B|CA}\rho^{\text{sep}}_{B|CA} + p_{C|AB}\rho^{\text{sep}}_{C|AB}$$

[Main Idea] Checking whether PPT mixture exists

It is well-known that separable states have Positive Partial Transpose (PPT). We denote such PPT states by $\rho^{\mathrm{ppt}}_{M|\bar{M}}$. Thus, we ask whether a state can be written as

$$\rho^{\mathrm{pmix}} = \sum_i p_i \rho^{\mathrm{ppt}}_{M_i|\bar{M}_i}$$

We call states of this form **PPT mixtures**.

Clearly, any biseparable state is a PPT mixture, so proving that a state is *not* a PPT mixture implies **genuine multipartite entanglement**.

[Recap] Positive Partial Transpose

For a bipartite state ρ_{AB} with matrix elements $\rho_{ij,kl} = \langle i, j|\rho|k, l\rangle$:

$$\rho^{T_A}_{ij,kl} = \rho_{kj,il}$$

A state has **Positive Partial Transpose (PPT)** if $\rho^{T_A} \geq 0$.

Quick Example: Bell state

$$\rho = |\phi^+\rangle\langle\phi^+| = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix} \to \rho^{T_A} = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Eigenvalues of ρ^{T_A}: $\left\{\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2}\right\}$
The **negative eigenvalue** proves the Bell state is entangled!

[Main Idea] Characterization via entanglement witnesses

Definition

An observable W is an **entanglement witness** for GME if:

- $\mathrm{Tr}(W\rho) \geq 0$ for all biseparable states ρ
- $\mathrm{Tr}(W\rho) < 0$ for some GME state ρ

Decomposable Witnesses

A witness is **decomposable** with respect to bipartition $M|\bar{M}$ if:

$$W = P_M + Q_M^{T_M}$$

where $P_M \geq 0$ and $Q_M \geq 0$.

A witness W is **fully decomposable** if for **all** bipartitions M:

$$W = P_M + Q_M^{T_M}, \quad P_M \geq 0, \quad Q_M \geq 0$$

[Main Idea] Fully Decomposable Witnesses

Theorem

Fully decomposable witnesses detect **exactly** the states that are **not PPT mixtures**.

If ρ is not a PPT mixture, then there exists a fully decomposable witness W that detects ρ.

$$\rho \notin \mathcal{S}_{\text{PPT-mix}} \implies \exists\, W \in \mathcal{W}_{\text{fd}} : \mathrm{Tr}(W\rho) < 0$$

where

- $\mathcal{S}_{\text{PPT-mix}}$ = set of PPT mixtures
- $\mathcal{W}_{\text{fd}}$ = set of fully decomposable witnesses
- $\mathrm{Tr}(W\rho) < 0$ = "W detects ρ"

This suggests that **if we find a fully decomposable W with $\mathrm{Tr}(W\rho) < 0$, the state is GME!**

[Example] Fully Decomposable Witnesses

Let's return to the Bell state in a two-qubit system. There is only one bipartition, $A|B$. A natural candidate witness for the Bell state is the **projector witness**: $W = \frac{1}{2}I - |\Phi^+\rangle\langle\Phi^+|$.

$$W = \frac{1}{2}I - |\Phi^+\rangle\langle\Phi^+| = \frac{1}{2}\begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix}$$

This entanglement witness detects the Bell state:

$$\mathrm{Tr} W|\Phi^+\rangle\langle\Phi^+| = -\frac{1}{2}$$

[Example] Fully Decomposable Witnesses

We need to check if $W = P + Q^{T_A}$ for some $P, Q \geq 0$.

We can simply choose:

$$P = 0, \qquad Q = W^{T_A} = \frac{1}{2}I - (|\Phi^+\rangle\langle\Phi^+|)^{T_A} = \frac{1}{2}\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

One can verify that $\mathrm{Tr}(W) = 1$, and $P, Q \geq 0$, confirming W is fully decomposable.

In general, finding such decompositions is non-trivial.

Semidefinite Program for GME Detection

$$\begin{aligned} \min_{W,P_M,Q_M} \quad & \mathrm{Tr}(W\rho) \\ \text{subject to} \quad & \mathrm{Tr}(W) = 1 \\ & W = P_M + Q_M^{T_M}, P_M \geq 0, Q_M \geq 0 \quad \forall M \end{aligned}$$

The free parameters are W and the operators P_M, Q_M for every bipartition M. If the minimum of $\mathrm{Tr}(W\rho)$ is negative, then ρ is not a PPT mixture and hence is **GME**.

This optimization takes the form of a semidefinite program (SDP). The global optimality of an SDP can be certified and the solution can be efficiently computed via interior-point methods.

Additional Strategies

- **Fully PPT witnesses:** Set $P_M = 0$ so that $W^{T_M} \geq 0$ for all M.
- **Observable basis:** Let $O = \{O_1, \cdots, O_k\}$ be a set of observables. Then parameterize $W = \sum_i \lambda_i O_i$

[Results] White Noise Tolerance

Consider mixed states(isotropic state): $\rho(p) = p \cdot \frac{I}{2^N} + (1-p) \cdot |\psi\rangle\langle\psi|$

Critical noise tolerance p_{tol}: Maximum p for which GME is still detected. A higher p_{tol} indicates stronger entanglement.

State $\|\psi\rangle$	**PPT mixtures**
GHZ_3	**0.571**
W_3	**0.521**
$\mathrm{Cluster}_4$	**0.615**
Dicke_4	**0.539**
GHZ_4	**0.500**

[Numerical Simulation] White Noise Tolerance

[Strategy] Modified SDP for Quantifying GME

This approach can also be used to quantify GME. If the trace normalization $\mathrm{Tr}(W) = 1$ is replaced by $0 \leq P_M \leq I$ and $0 \leq Q_M \leq I$, the negative witness expectation value becomes a multipartite entanglement monotone. Note that this quantity equals the **negativity** in the bipartite case.

$$\begin{aligned} -\min_{W, P_M, Q_M} \quad & \mathrm{Tr}(W\rho) \\ \text{subject to} \quad & \mathrm{Tr}(W) = 1 \\ & W = P_M + Q_M^{T_M}, 0 \leq P_M, Q_M \leq I \quad \forall M \end{aligned}$$

Properties of the Monotone $\mathcal{N}(\rho) = -\min \mathrm{Tr}(W\rho)$

- **Vanishes on biseparable states:** $\mathcal{N}(\rho^{\text{bs}}) = 0$
- **Convex:** $\mathcal{N}(\sum_i p_i \rho_i) \leq \sum_i p_i \mathcal{N}(\rho_i)$
- **LOCC monotone:** Does not increase under local operations and classical communication

[Numerical Result] $\mathcal{N}(\rho)$ on the two-qubit isotropic state

[Numerical Result] Quantifying GME

Summary

Pros

- The framework is systematic and complete.
- It provides a unified SDP-based approach for detecting GME.
- No false negatives within the PPT mixture class.
- The entanglement monotone not only detects but also quantifies GME.

Summary

Cons

- Generally needs full density matrix ρ — expensive quantum state tomography required
- No Closed-Form Solution. SDP optimization (convex optimization) is required for each state.
    - Time Complexity: $\mathcal{O}(\sqrt{m}D^3(m + D^2))$
    - Space Complexity: $\mathcal{O}(mD^2 + D^4)$

  where D is the Hilbert space dimension and m is the number of constraints.
- Not Complete for All Entanglement. Only detects states outside PPT mixtures; PPT entangled states are missed

References

Main Paper

- Jungnitsch, Bastian, Tobias Moroder, and Otfried Gühne. "Taming multiparticle entanglement." Physical review letters 106.19 (2011): 190502.

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Software

- CVXPY: https://www.cvxpy.org/

Thank You!

Questions & Discussion